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Data Science

Health Monitoring System

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Project Guide

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Sustainable Development Goal (SDG) Mapping

- SDG 3 focuses on good health and well-being. A health monitoring system supports this by tracking vital signs and enabling early disease detection. It allows remote monitoring and improves access to healthcare. Overall, it helps in better patient care and enhances quality of life
- SDG 9 focuses on building strong infrastructure, promoting innovation, and supporting modern industries. It encourages the use of advanced technologies like IoT, AI, and digital systems. In a health monitoring system, this goal is achieved by using innovative technology to improve healthcare services and efficiency.

1. Introduction

Problem Identified

•Many people, especially elderly patients, find it difficult to regularly monitor health parameters like blood pressure, blood sugar, and heart rate.

•Health records are often not properly maintained, making it hard to track health trends.

•Patients sometimes forget to take medicines on time.

Solution Proposed

•Develop a mobile-based Health Monitoring System to record daily health data.

•Provide graphs and alerts to detect abnormal health values.

•Include medicine reminders for timely medication.

•Store health data securely using SQLite / Firebase database.

2. Objectives

1. To develop a mobile-based Health Monitoring System for tracking daily health parameters.

2. To monitor important health data such as blood pressure, blood sugar, and heart rate.

3. To analyze health data and detect abnormal trends at an early stage.

4. To provide alerts and notifications when abnormal health values are detected.

5. To remind users to take medicines on time through a reminder system.

6. To maintain secure digital records of users' health data for future reference.

3. Scope

- Can be applied in personal health monitoring for daily tracking of BP, blood sugar, and heart rate.
- Can be useful to elderly patients and people with chronic diseases who need regular health monitoring.
- Can be used in home healthcare environments for maintaining digital health records.
- Can be useful in preventive healthcare by helping detect abnormal health trends early.
- Can be applied in the healthcare domain for improving medication adherence through reminders.
- Can be extended in future by integrating with wearable devices and hospital health systems.

4. Feature /Functionality

Feature 1: Smart Health Monitoring

The application allows users to record important health parameters such as blood pressure, blood sugar, and heart rate in one place. It helps users track their daily health condition easily.

Feature 2: Intelligent Alerts & Medicine Reminder

The system automatically detects abnormal health values and sends alerts to the user. It also provides medicine reminders to ensure timely medication and better treatment adherence.

Feature 3: Health Trend Visualization

The application analyzes stored health data and displays it through graphs and reports. This helps users understand health trends and take preventive action early.

5. Outcome of Project

1. User Registration & Login – Users can create an account and log in securely to access the system.

2. Health Dashboard – Displays key health parameters like heart rate, blood pressure, and sugar level.

3. Health Data Entry – Users can enter and store daily health records for tracking.

4. Health Trend Analysis – The system shows health trends using daily, weekly, and monthly graphs.

5. Alerts & Medicine Reminder – Provides alerts for abnormal values and reminders for medicines.

6.Technology Stack

Front-End (GUI): Flutter

• Designed a simple and attractive user interface for the Health Monitoring System with screens like login, dashboard, vitals, medicines, alerts, and PDF reports.

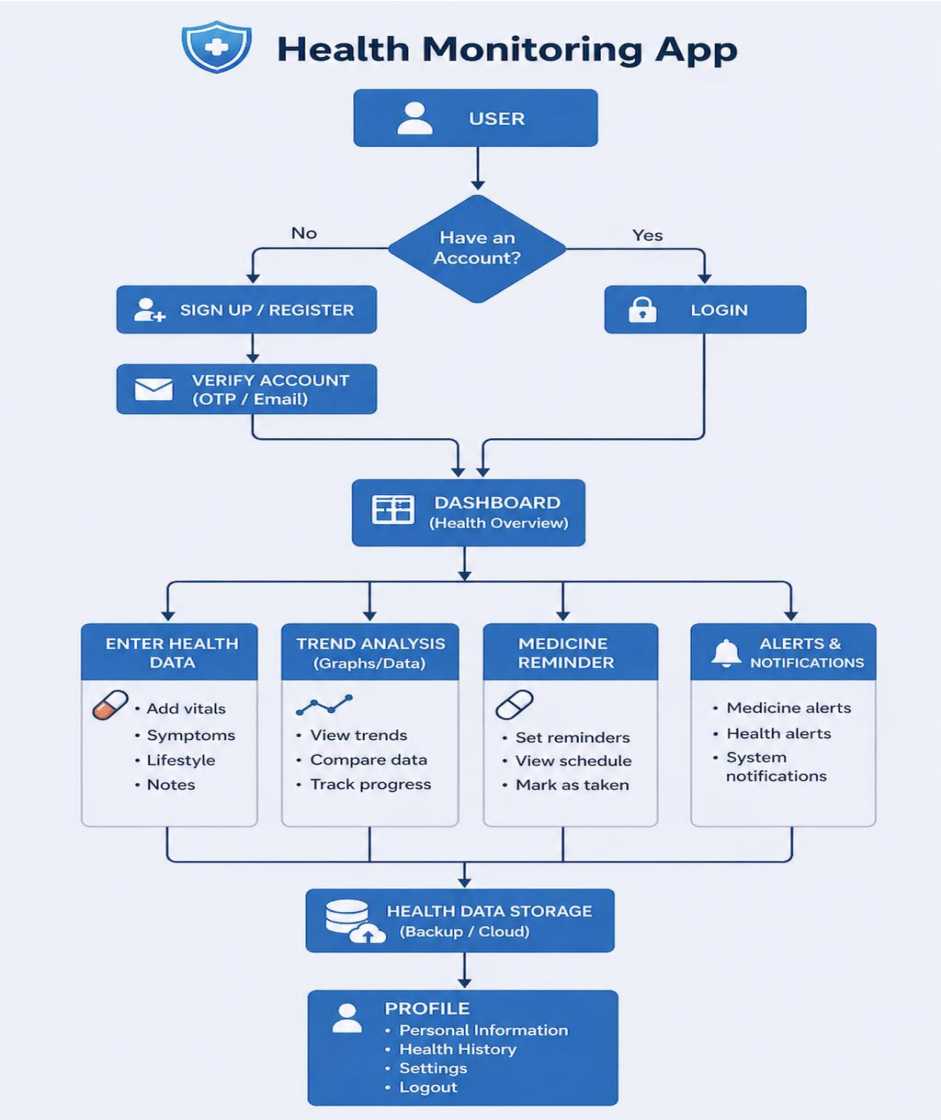
• Implemented features like real-time health tracking, graphical analysis, reminders, and easy navigation so users can manage their health efficiently.

Back-End (Database): Firebase

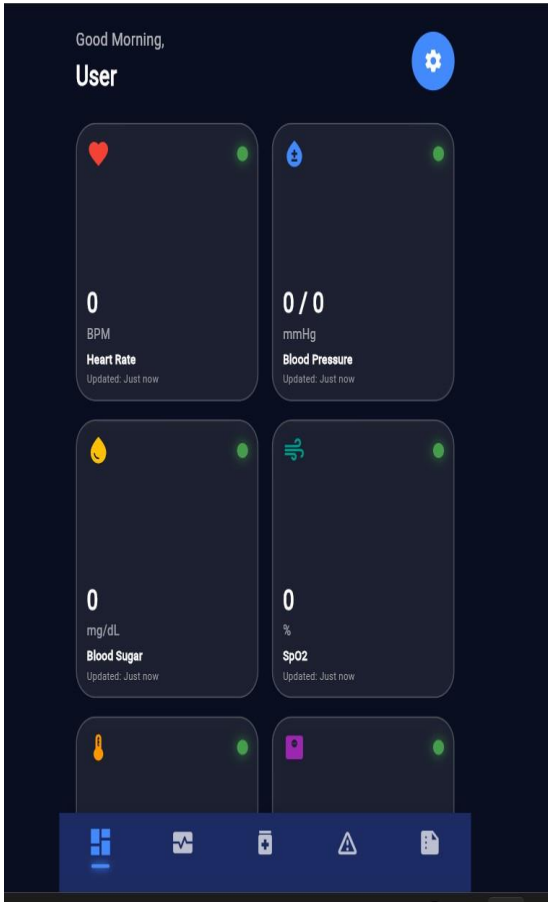
• Developed a secure backend using Firebase services to store user health records, medicine schedules, and alerts safely in the cloud.

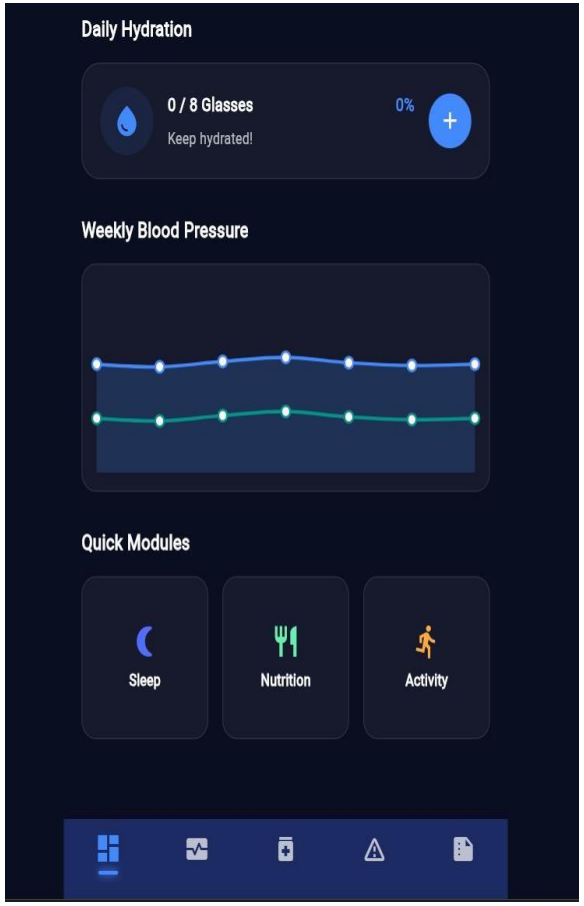
• Implemented Firebase Authentication, Cloud Firestore, and cloud storage to ensure secure login, data management, and proper functioning of the system.

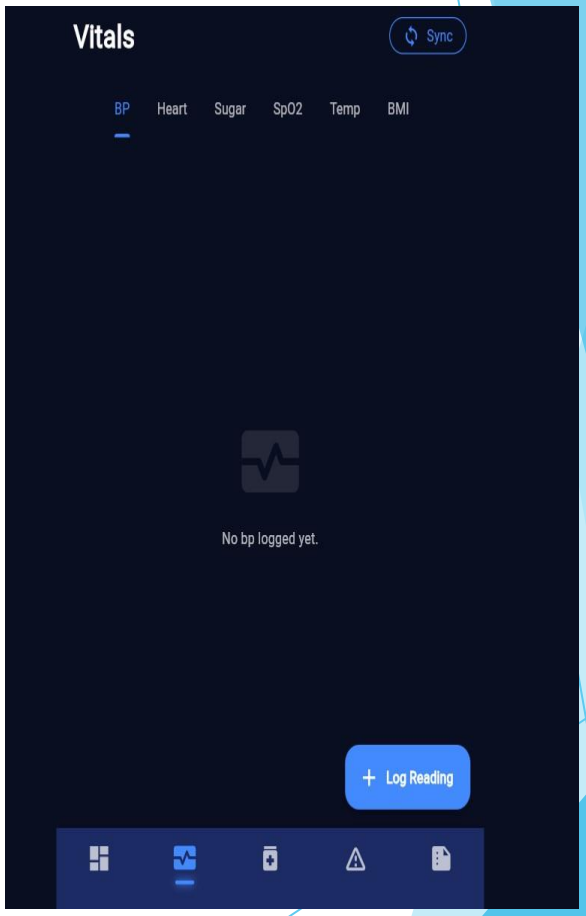
7. Block Diagram

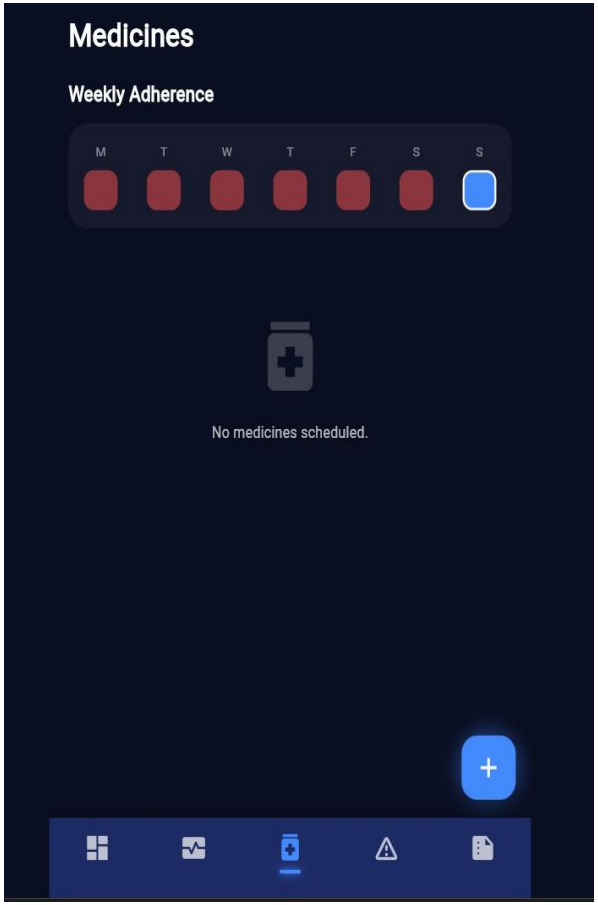


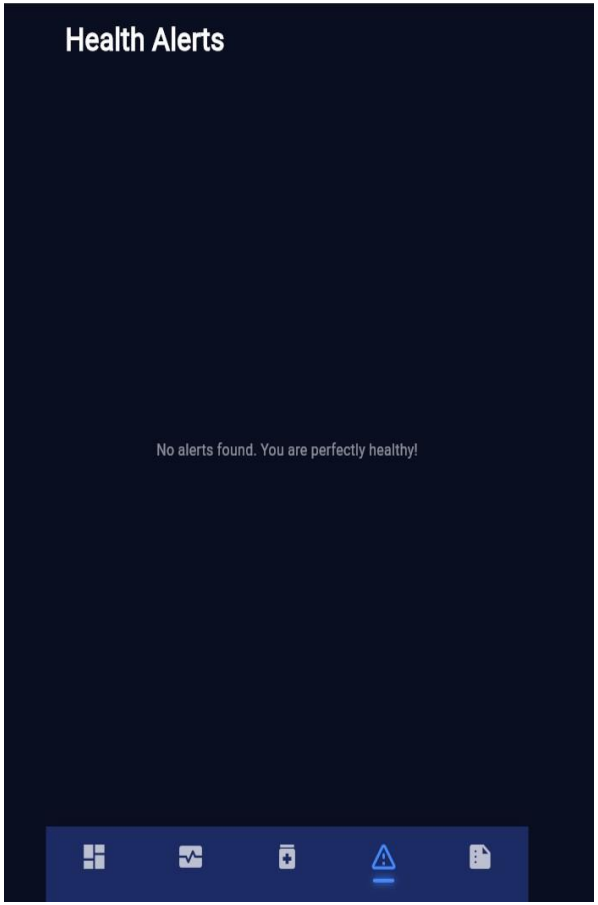
8.Frontend

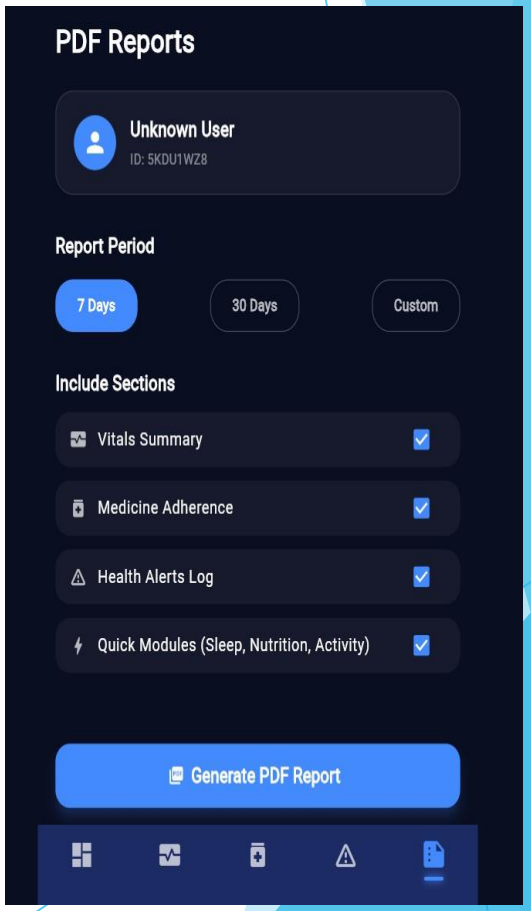

Good Morning,
User
0
BPM
Heart Rate
Updated: Just now
0 / 0
mmHg
Blood Pressure
Updated: Just now
0
mg/dL
Blood Sugar
Updated: Just now
0
%
SpO2
Updated: Just now


Daily Hydration
0 / 8 Glasses
Keep hydrated!
0%
Weekly Blood Pressure
Quick Modules
Sleep
Nutrition
Activity

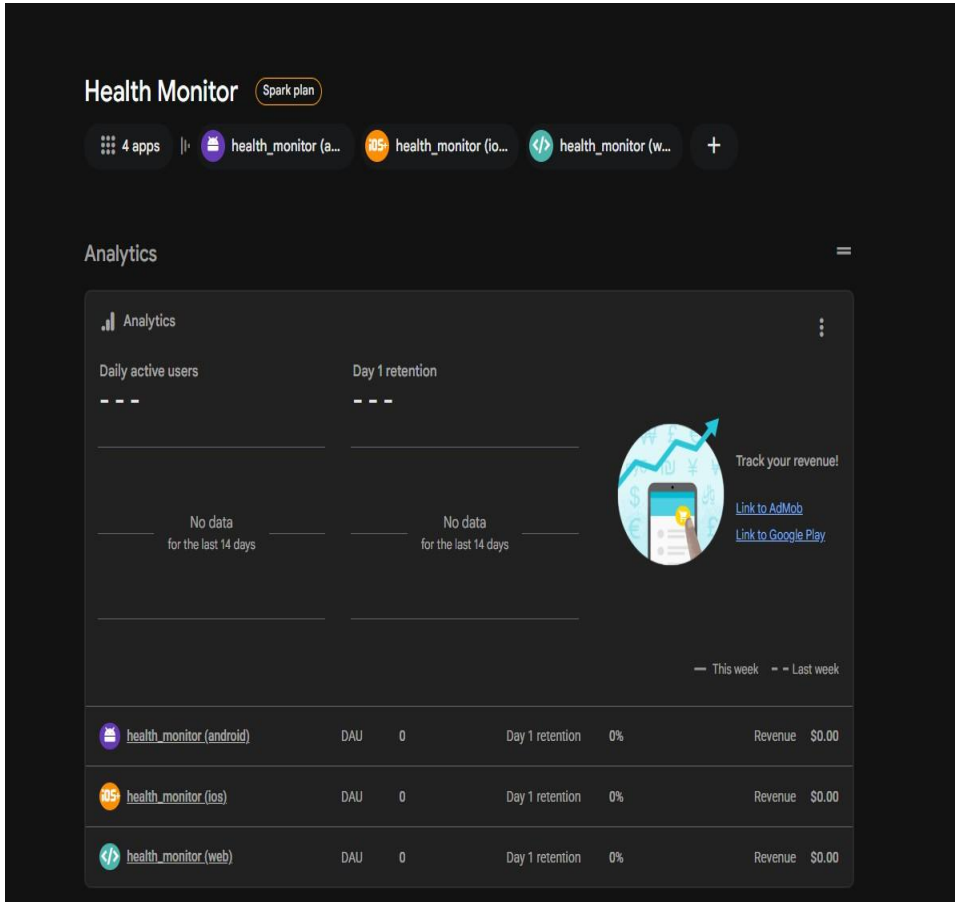

Vitals
Sync
BP
Heart
Sugar
SpO2
Temp
BMI
No bp logged yet.
Log Reading

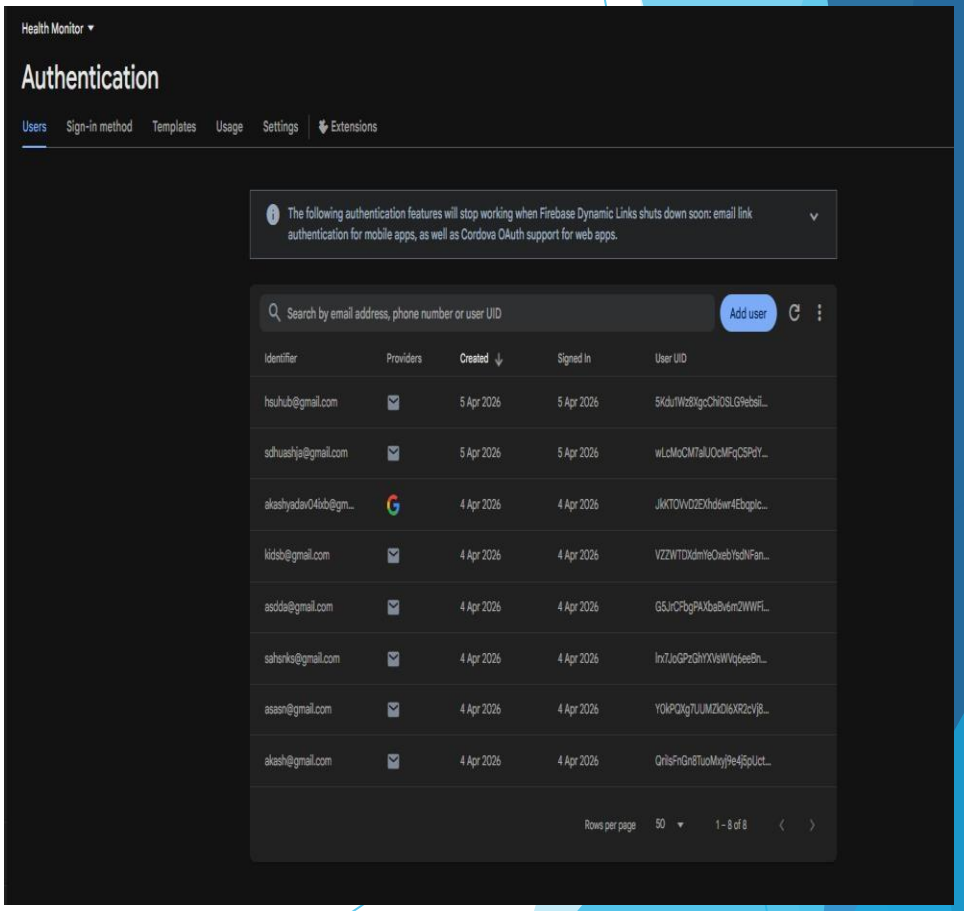

Medicines
Weekly Adherence
M T W T F S S
No medicines scheduled.


Health Alerts
No alerts found. You are perfectly healthy!


PDF Reports
Unknown User
ID: 5KDU1WZ8
Report Period
7 Days
30 Days
Custom
Include Sections
Vitals Summary
Medicine Adherence
Health Alerts Log
Quick Modules (Sleep, Nutrition, Activity)
Generate PDF Report

9.Backend


Health Monitor
Spark plan
4 apps
health_monitor (a...
health_monitor (io...
health_monitor (w...
Analytics
Analytics
Daily active users
- - -
Day 1 retention
- - -
No data
for the last 14 days
No data
for the last 14 days
Track your revenue!
Link to AdMob
Link to Google Play
This week
Last week
health_monitor (android) DAU 0 Day 1 retention 0% Revenue $0.00
health_monitor (ios) DAU 0 Day 1 retention 0% Revenue $0.00
health_monitor (web) DAU 0 Day 1 retention 0% Revenue $0.00


Health Monitor
Authentication
Users
Sign-in method
Templates
Usage
Settings
Extensions
The following authentication features will stop working when Firebase Dynamic Links shuts down soon: email link authentication for mobile apps, as well as Cordova OAuth support for web apps.
Search by email address, phone number or user UID
Add user
Identifier
Providers
Created
Signed In
User UID
hsuhub@gmail.com 5 Apr 2026 5 Apr 2026
sdhuashja@gmail.com 5 Apr 2026 5 Apr 2026
akashyadav04ixb@gm... 4 Apr 2026 4 Apr 2026
kidsb@gmail.com 4 Apr 2026 4 Apr 2026
asdda@gmail.com 4 Apr 2026 4 Apr 2026
sahsnks@gmail.com 4 Apr 2026 4 Apr 2026
asasn@gmail.com 4 Apr 2026 4 Apr 2026
akash@gmail.com 4 Apr 2026 4 Apr 2026
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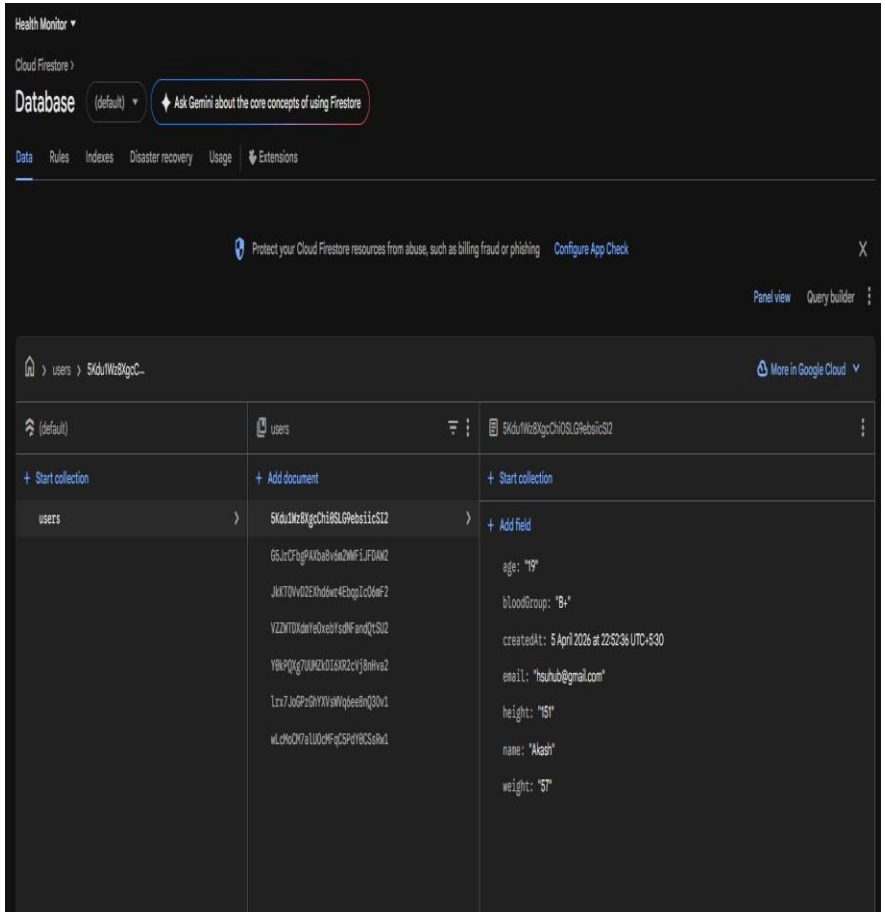
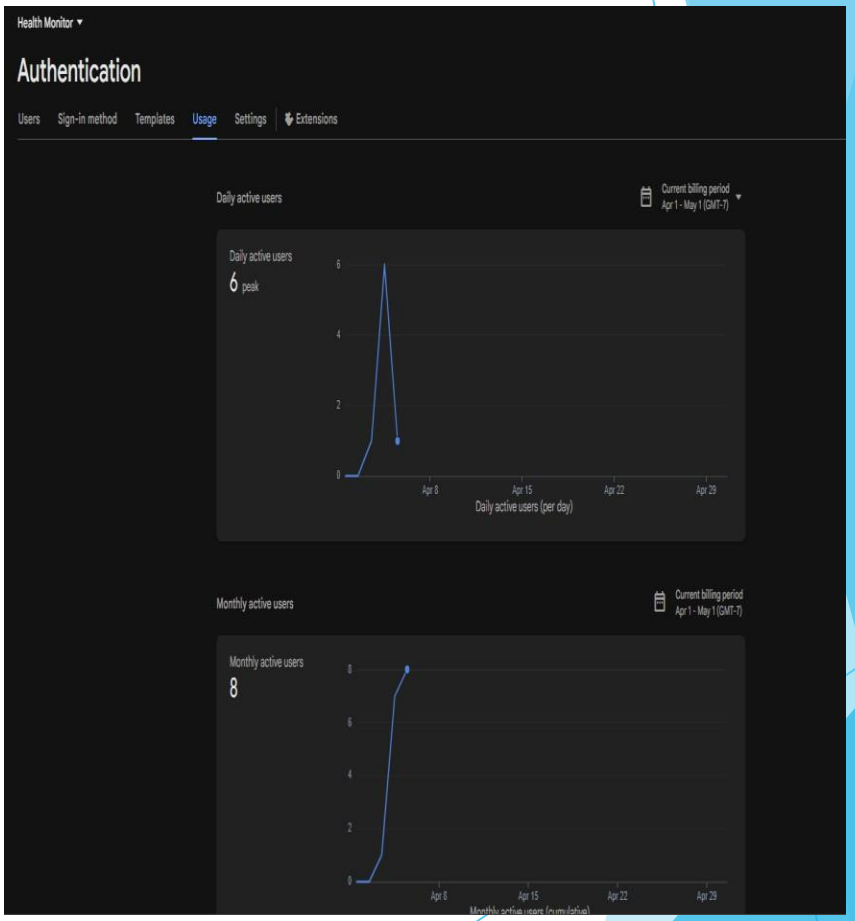
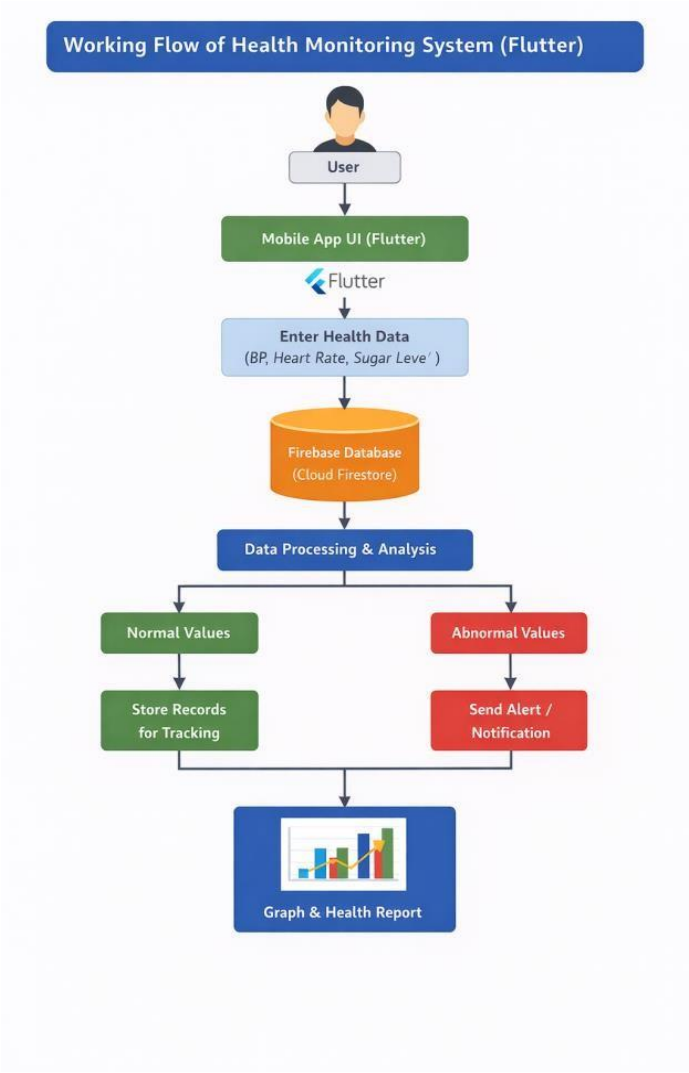

Health Monitor
Cloud Firestore
Database
(default)
Ask Gemini about the core concepts of using Firestore
Data
Rules
Indexes
Disaster recovery
Usage
Extensions
Protect your Cloud Firestore resources from abuse, such as billing fraud or phishing
Configure App Check
Panel view
Query builder
users
More in Google Cloud
(default)
users
Start collection
Add document
Start collection
Add field
age: "19"
bloodGroup: "B+"
createdAt: 5 April 2026 at 22:52:36 UTC+5:30
email: "hsuhub@gmail.com"
height: "151"
name: "Akash"
weight: "57"


Health Monitor
Authentication
Users
Sign-in method
Templates
Usage
Settings
Extensions
Daily active users
Current billing period
Apr 1 - May 1 (GMT-7)
Daily active users
6 peak
Apr 8
Apr 15
Apr 22
Apr 29
Daily active users (per day)
Monthly active users
Current billing period
Apr 1 - May 1 (GMT-7)
Monthly active users
8
Apr 8
Apr 15
Apr 22
Apr 29

10.Workflow


Working Flow of Health Monitoring System (Flutter)
User
Mobile App UI (Flutter)
Flutter
Enter Health Data
(BP, Heart Rate, Sugar Level)
Firebase Database
(Cloud Firestore)
Data Processing & Analysis
Normal Values
Abnormal Values
Store Records
for Tracking
Send Alert /
Notification
Graph & Health Report

Thank You...!!